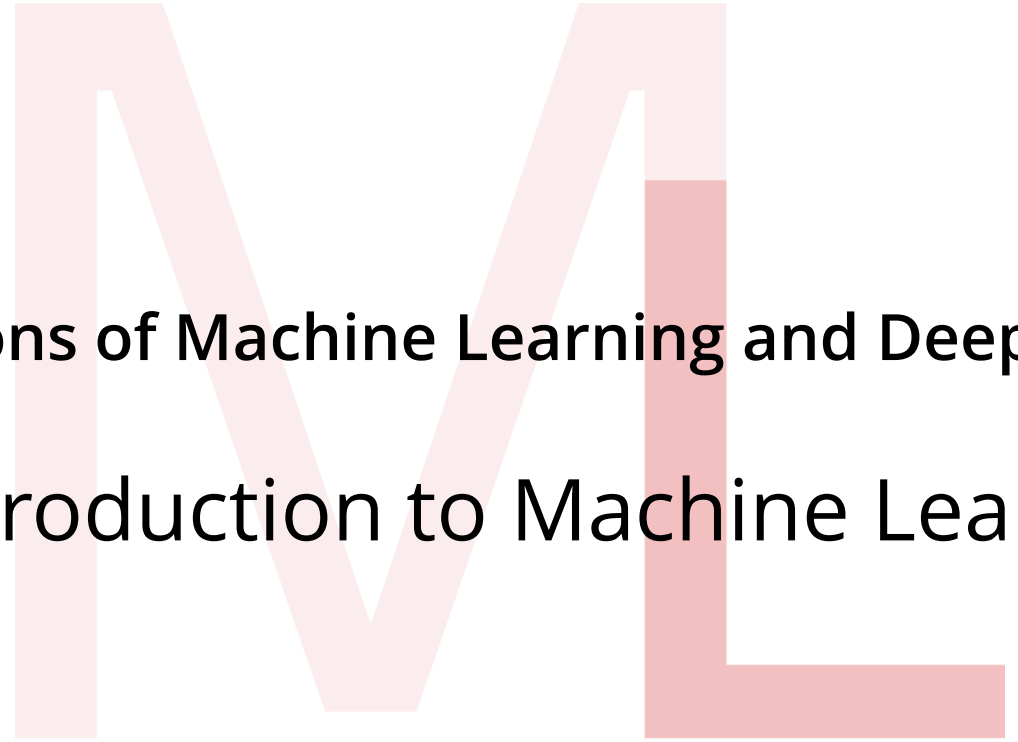
Foundations of Machine Learning and Deep Learning

1. Introduction to Machine Learning



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Outline

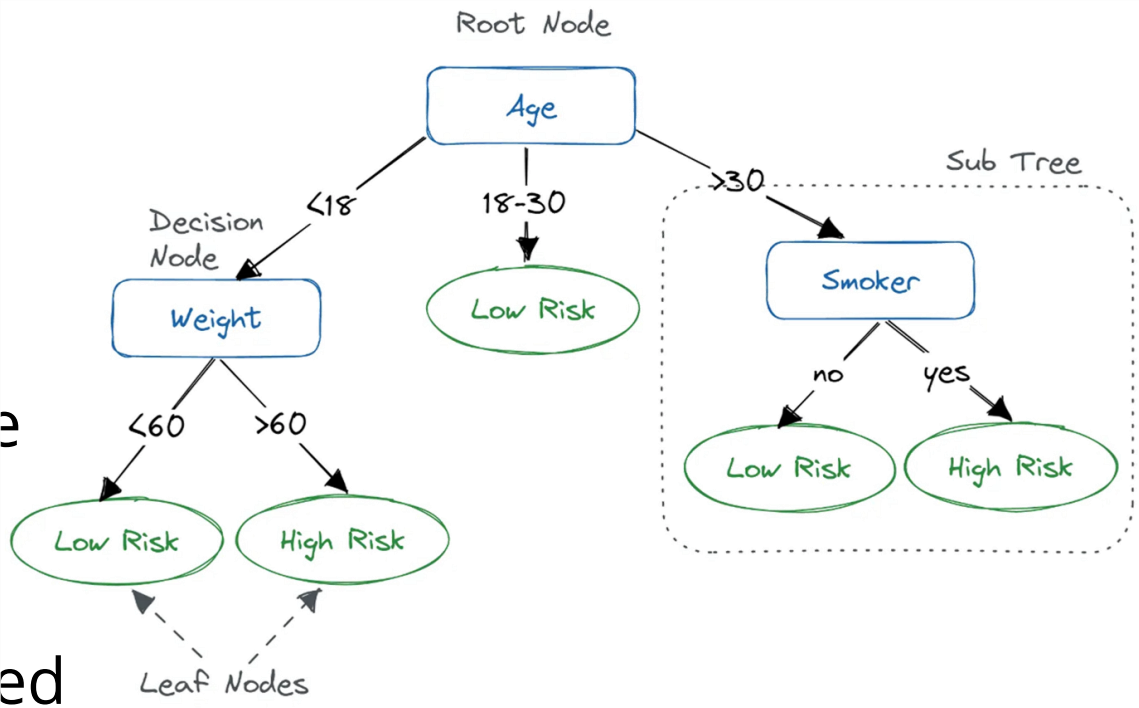
- What is Machine Learning?
- Notation & Core Concepts
- Learning paradigms
- Classical Models
  - K nearest neighbors
  - Naive Bayes
  - Decision trees
  - Ensemble models
- Model evaluation and selection
- Clustering

Classical Models – Decision trees



Classical Models – Decision trees

- **Decision trees:** Interpretable models for classification and regression
  - A tree-structured model that makes decision following a sequence of rules
  - Various types of nodes:
    - The root node is the start of the tree
    - Each internal node represents a decision on a feature
    - Each leaf node represents a predicted output (class or numeric value)
  - Easy to interpret, visualize and explain



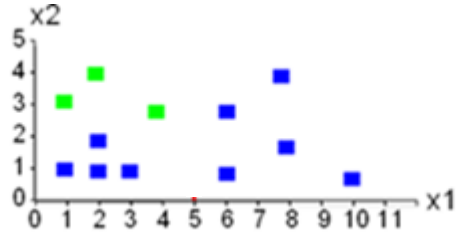
Classical Models – Decision trees

- **Construction of a decision tree**
  - The idea is to recursively split the data to create pure subsets until a stopping condition is met
  - The construction is greedy
  - Stopping criteria:
    - All simples in the node belong to the same class
    - Maximum tree depth is reached
    - Minimum number of samples per node is reached
    - No feature provides any gain

```
function build_tree(data):
    if all samples have the same class:
        return a leaf node with that class
    if stopping condition is met:
        return a leaf node with majority class
    else:
        for each feature:
            evaluate split quality (e.g., info gain, Gini)
        choose best feature and threshold
        split data into left/right (or multiway) subsets
        create an internal node
        recursively build left and right subtrees
        return the node
```

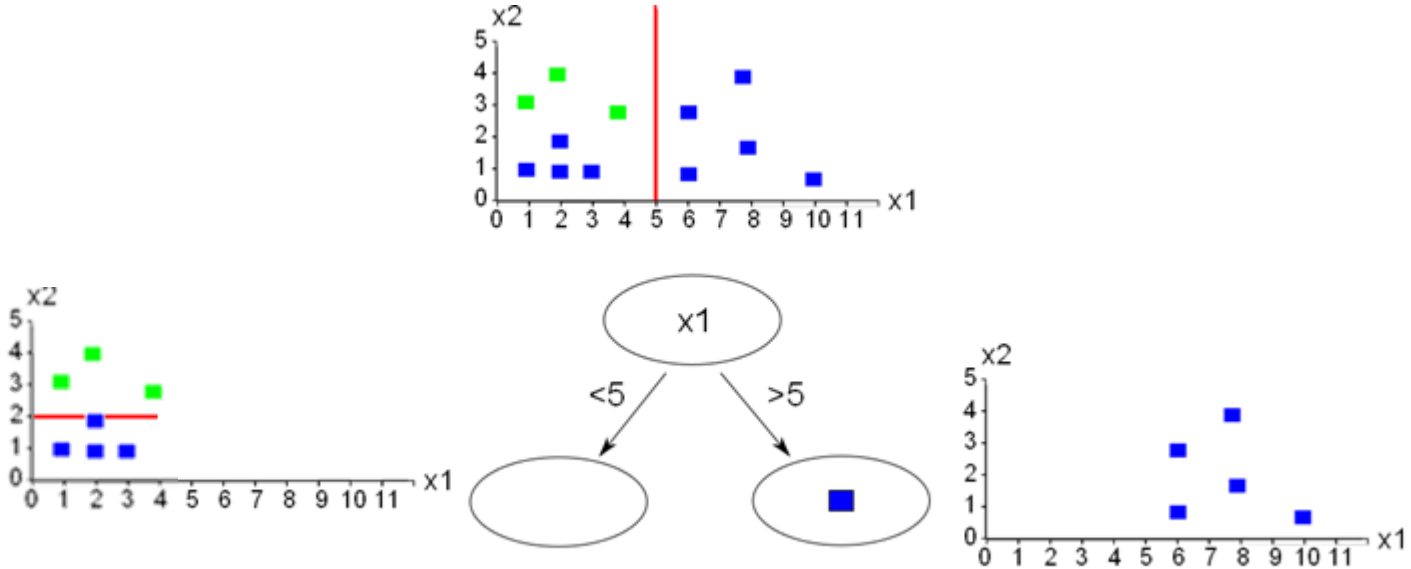
Classical Models – Decision trees

- **Recursive partitioning**



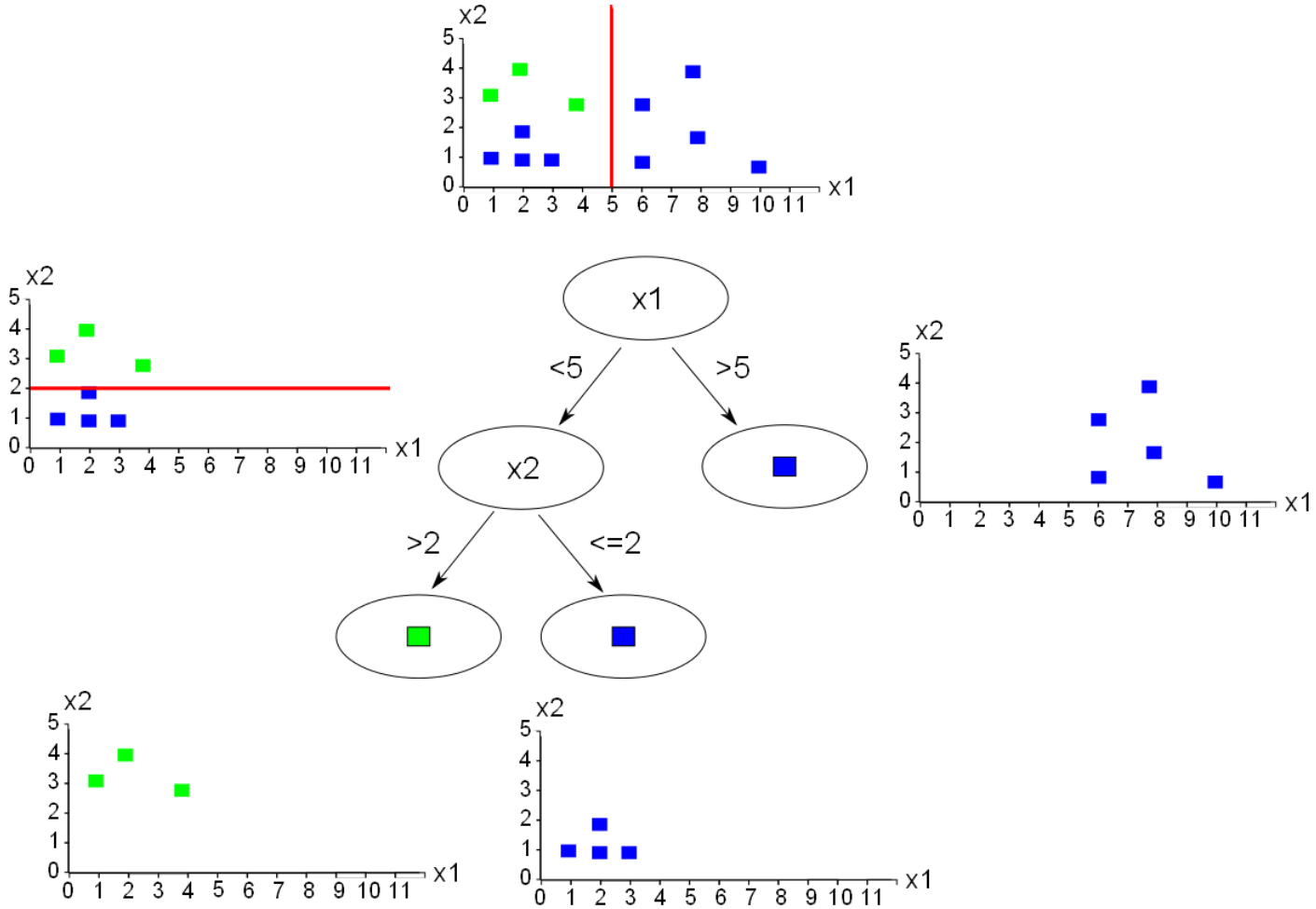
Classical Models – Decision trees

- **Recursive partitioning**



Classical Models – Decision trees

- **Recursive partitioning**



Classical Models – Decision trees

- There are **different decision tree algorithms with different splitting strategies**
  - ID3
    - Information Gain
  - C4.5
    - Gain Ratio
  - CART
    - Gini Impurity

Classical Models – Decision trees

- **ID3 (Iterative dichotomiser 3)**
  - Only works with categorical features
  - No pruning
  - Uses Information Gain (based on entropy) as the split criterion:
    - Idea: Measuring how much uncertainty (entropy) is reduced by the split
    - Entropy:

$$H(S) = -\sum_{i=1}^{c} p_i \log_2 p_i$$

      where
      - S is the set of samples at the current node
      - p_i is the proportion of examples of class *i* in the current node
    - The entropy is 0 if all samples are of the same class
    - It is maximal (1 for binary classification) if classes are evenly split

Classical Models – Decision trees

- **ID3 (Iterative dichotomiser 3)**
  - Information Gain:

$$IG(S, A) = H(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

where

- S is the set of samples at the current node
- A is a feature we might split on, and v is a value that it can take
- S_v is the subset of samples in S where feature $A = v$

  - It measures how much the overall impurity is reduced after splitting on A

- **ID3 (Iterative dichotomiser 3) – Example**
  - What is the first split?

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Classical Models – Decision trees

- **ID3 (Iterative dichotomiser 3) – Example**
  - What is the first split?

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
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7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

1. Compute the entropy of the entire dataset

Count(Yes) = 9
Count(No) = 5

$$H(S) = -\left(\frac{9}{14}\log_2\frac{9}{14} + \frac{5}{14}\log_2\frac{5}{14}\right) \approx 0.940$$

2. Evaluate the split on each feature

Classical Models – Decision trees

- **ID3 (Iterative dichotomiser 3) – Example**
  - What is the first split?

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
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9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Count(Yes) = 9
Count(No) = 5

1. Entropy of entire dataset $H(S) \approx 0.940$
2. Evaluate the split on each feature
   - **Outlook**

Outlook	Count	Play=Yes	Play=No	Entropy
Sunny	5	2	3	0.971
Overcast	4	4	0	0.000 (pure)
Rain	5	3	2	0.971

$$H(S, \text{Outlook}) = \frac{5}{14} \cdot 0.971 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.971 = 0.693$$

$$IG(S, \text{Outlook}) = 0.940 - 0.693 = \boxed{0.247}$$

Classical Models – Decision trees

- **ID3 (Iterative dichotomiser 3) – Example**
  - What is the first split?

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
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12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Count(Yes) = 9
Count(No) = 5

1. Entropy of entire dataset $H(S) \approx 0.940$
2. Evaluate the split on each feature

Feature	Information Gain
Outlook	**0.247**
Humidity	0.151
Wind	0.048
Temperature	0.029

3. **Compare and split**: In this case, outlook has the highest IG

- **ID3 (Iterative dichotomiser 3) – Example**
  - What happens if Day is a valid attribute for the tree?

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
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12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Classical Models – Decision trees

- **Split Info**
  - Information about the division of the attribute, independent from the objective variable (class)

$$\text{SplitInfo}(A) = -\sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \log_2 \frac{|S_v|}{|S|}$$

    - If an attribute splits the data into two equal halves (2 branches), its SplitInfo is low ($\log_2(2) = 1$)
    - If an attribute splits a dataset of 100 instances into 100 branches (1 instance per branch), its SplitInfo is extremely high ($\log_2(100) = 6{,}64$)

Classical Models – Decision trees

- **C4.5**
  - It is an extension of ID3
    - Uses **Gain Ratio** instead of Information Gain

- Gain Ratio:

$$\text{GainRatio}(S, A) = \frac{IG(S, A)}{\text{SplitInfo}(A)}$$

SplitInfo measures how broadly and evenly the attribute A splits the data and **GainRatio** penalizes over-fragmentation

Classical Models – Decision trees

- **C4.5**
  - It is an extension of ID3
    - Uses **Gain Ratio** instead of Information Gain
    - Handles both categorical and continuous features
    - Applies pruning techniques (to reduce overfitting)

Classical Models – Decision trees

- **C4.5**
  - Continuous features:
    - It evaluates splits of the form $\text{Feature} \leq t \quad \text{vs.} \quad \text{Feature} > t$
    - Tries some thresholds *t*
      - Midpoints between consecutive values where the class label changes
    - The threshold *t* that maximizes the Gain Ratio is chosen
      - The split for continuous features is **binary**

Classical Models – Decision trees

- **C4.5**
  - Continuous features:

Instance	D1	D2	D3	D4	D5	D6
Temperatura (°C)	15	18	21	24	27	30
Play Tennis	No	No	Sí	Sí	No	Sí

t=19,5

t=25,5

t=28,5

Classical Models – Decision trees

- **C4.5**
  - Continuous features:

Instance	D1	D2	D3	D4	D5	D6
Temperature (°C)	15	18	21	24	27	30
Play Tennis	No	No	Yes	Yes	No	Yes

- t=19,5
  - Left (2 No); Right (3 Yes, 1 No) → $H = \frac{2}{6}0 + \frac{4}{6}0{,}811 = 0{,}541$ → $IG = 0{,}459$
- t=25,5
  - Left (2 Yes, 2 No); Right (1 Yes, 1 No) → $H = \frac{4}{6}1 + \frac{2}{6}1 = 1$ → $\mathrm{IG} = 0$
- t=28,5
  - Left (2 Yes, 3 No); Right (1 Yes) → $H = \frac{5}{6}0{,}971 + \frac{1}{6}0 = 0{,}809$ → $IG = 0{,}191$

Classical Models – Decision trees

- **C4.5**
  - Continuous features:

Instance	D1	D2	D3	D4	D5	D6
Temperature (°C)	15	18	21	24	27	30
Play Tennis	No	No	Yes	Yes	No	Yes

- t=19,5 (2 left, 4 right)
  - $SplitInfo = \frac{2}{6}\log_2\frac{2}{6} + \frac{4}{6}\log_2\frac{4}{6} = 0{,}918$
- t=25,5 (4 left, 2 right)
  - $SplitInfo = \frac{4}{6}\log_2\frac{4}{6} + \frac{2}{6}\log_2\frac{2}{6} = 0{,}918$
- t=28,5 (5 left, 1 right)
  - $SplitInfo = \frac{5}{6}\log_2\frac{5}{6} + \frac{1}{6}\log_2\frac{1}{6} = 0{,}650$

Classical Models – Decision trees

- **C4.5**
  - Continuous features:

Instance	D1	D2	D3	D4	D5	D6
Temperature (°C)	15	18	21	24	27	30
Play Tennis	No	No	Yes	Yes	No	Yes

- t=19,5 → $IG = 0{,}459$ → $SplitInfo = 0{,}918$
  - $GainRatio = 0{,}5$
- t=25,5 → $\mathrm{IG} = 0$ → $SplitInfo = 0{,}918$
  - $GainRatio = 0$
- t=28,5 → $IG = 0{,}191$ → $SplitInfo = 0{,}650$
  - $GainRatio = 0{,}294$

- **C4.5**
  - Pruning: pessimistic pruning (after tree creation)
    - Training error based evaluation (optimistic, measured in the training set). Sum of the error of every leaf node

$$TE = \sum_{t \in Leaves} \frac{e_t \quad (wrong\ classified\ instances\ in\ leaf\ t)}{N_t \qquad (number\ of\ instances\ in\ leaf\ t)}$$

    - Pessimistic error: add penalty of 0,5 extra errors per leaf node

$$PE = \frac{TE + 0{,}5 \cdot |L| \qquad (|L| is\ the\ number\ of\ leaves)}{N \qquad (used\ to\ transform\ into\ percentaga)}$$

    - Replace a subtree with a single leaf if the leaf's pessimistic error is within 1 Standard Error of the original subtree error

$$SE = \sqrt{N \cdot PE \cdot (1 - PE)}$$

Classical Models – Decision trees

- **CART (Classification and Regression Trees)**
  - Handles continuous and categorical features
  - Supports **binary** splits only (each node always splits into 2)
  - Uses cost-complexity pruning
  - Works for both classification and regression

  - Splitting criterion: minimizing **Gini impurity** (the lower the better)

$$G(S) = 1 - \sum_{i=1}^{c} p_i^2$$

where p_i is the proportion of class i in S

  - Similar to entropy, but computationally faster

Classical Models – Decision trees

- **How is GINI used?**
  1. For each candidate split:
     - Partition the dataset S into two child subsets: S_1 , S_2 (or S_{left} , S_{right})
  2. Compute the Gini impurity of each subset

$$G(S_j) = 1 - \sum_{i=1}^{c} p_{ij}^2$$

     where p_{ij} is the proportion of class i in subset S_j
  3. Compute the weighted average Gini after the split:

$$G_{\text{split}} = \sum_{j} \frac{|S_j|}{|S|} G(S_j)$$

  4. The **best split** is the one with the lowest G_{split}

- **The regression case: Minimizing the variance**
  - At each node, the algorithm chooses the split that minimizes the sum of squared errors (SSE) or variance of the target:

$$\text{Variance}(S) = \frac{1}{|S|} \sum_{i=1}^{|S|} (y_i - \bar{y})^2$$

where

- y_i is target value for sample i
- $\bar{y}$ is the mean of the target values in node S

Classical Models – Decision trees

- **Summary**
  - Decision trees are interpretable
  - They tend to overfit
  - All algorithms build trees by splitting nodes to reduce impurity
  - C4.5 and CART both support continuous features and pruning
  - CART is the most used in real-world libraries
    - e.g., scikit-learn, XGBoost base learners

Feature	ID3	C4.5	CART
Split Criterion	Information Gain	Gain Ratio	Gini Impurity / Variance
Handles Numeric Data	No	Yes	Yes
Split Type	Multiway (categorical)	Multiway (cat), Binary (num)	Binary only
Target Type	Classification only	Classification only	Classification & Regression
Pruning	No	Pessimistic-pruning	Cost-complexity pruning